

Dynamics of System of Particles

If all parts of an object does not move in exactly the same way, then this object must be treated as a system of particles. (what is this?) X

Center of mass: Simplifies the understanding and analysis of the motion, stability, and interactions of objects and systems.

e.g.  : all parts of the block moves exactly in the same way. So, we can simply think that as if all its mass was concentrated at its center of mass, and as if that "concentrated" "point particle" was moving.

e.g.



Earth rotates around sun but it also rotates around itself. When it comes to describe its motion around the sun, we don't care its own rotation about its axis. We can take its C.O.M into consideration. Yet, when it comes to explain the, say, "tidal waves", we need to take care of its rotation about its axis.

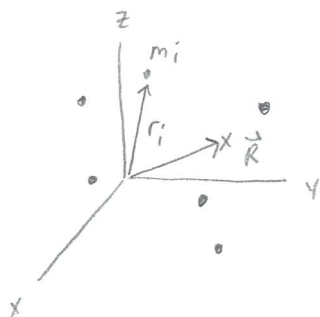
→ As a simplification, we can simply put our frame of reference at its C.O.M and try to explain its rotation from that point of view.

e.g. Collision and momentum:

* In collisions between objects, understanding the center of mass helps determine the conservation of momentum. The behavior of the system before and after the collision can be analyzed based on how C.O.M behaves.

e.g. Balance and stability

Discrete System of Particles:



$$\vec{R} = \frac{1}{M} \sum_i m_i \vec{r}_i$$

$$M = \sum_i m_i ; \text{ total mass of the system.}$$

$$x_{cm} = \frac{1}{M} \sum_i m_i x_i$$

$$\vec{R} = x_{cm} \hat{i} + y_{cm} \hat{j} + z_{cm} \hat{k}$$

$$y_{cm} = \frac{1}{M} \sum_i m_i y_i$$

$$z_{cm} = \frac{1}{M} \sum_i m_i z_i$$

Ex. A system of particles consists of 3 masses $m_A = 0.5 \text{ kg}$, $m_B = 2 \text{ kg}$ and $m_C = 5 \text{ kg}$ located at $\vec{r}_A = (-3, 1, 2) \text{ m}$, $\vec{r}_B = (0, 1, 2) \text{ m}$, $\vec{r}_C = (-1, 3, 0) \text{ m}$ respectively. Find the position vector of the COM of the system.

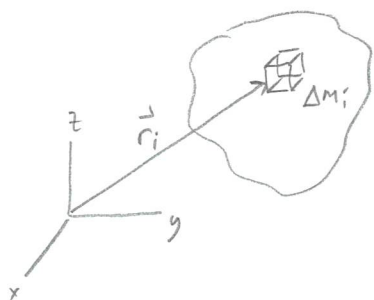
$$M = 7.5 \text{ kg} \quad x_{cm} = \frac{1}{7.5} \{ 0.5 \times (-3) + 2 \times 0 + 5 \times (-1) \} = \frac{1}{7.5} (-1.5 - 5) = -\frac{6.5}{7.5} \text{ m}$$

$$y_{cm} = \dots$$

$$z_{cm} = \dots$$

Continuous System of Particles

A continuous system of particles is a system consisting of a large number of particles separated by very small distances.



$$\vec{R} \approx \frac{1}{M} \sum_i \vec{r}_i \Delta m_i$$

$$\vec{R} = \lim_{\Delta m_i \rightarrow 0} \frac{1}{M} \sum_i \vec{r}_i \Delta m_i = \frac{1}{M} \int \vec{r} dm$$

In solving problems it is common to use density functions:

Volume density: $\rho(\vec{r})$: mass per unit volume. : can be a function of position.

$$M = \int \rho(\vec{r}) dV, \quad \vec{R} = \frac{\int \vec{r} \rho dV}{M}$$

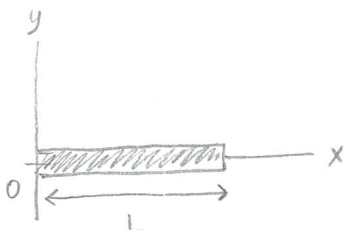
Surface density: $\sigma(\vec{r})$: mass per unit area

$$M = \int \sigma(\vec{r}) dA, \quad \vec{R} = \frac{\int \vec{r} \sigma dA}{M}$$

Linear density: $\lambda(\vec{r})$: mass per unit length

$$M = \int \lambda(\vec{r}) dl, \quad \vec{R} = \frac{\int \vec{r} \lambda dl}{M}$$

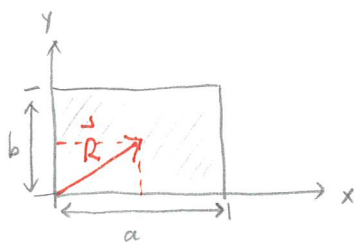
Ex:



(ideally linear)
A thin rod of length $L = 2\text{m}$ has a linear density that increases with x according to the expression $\lambda(x) = (2x-1) \frac{\text{kg}}{\text{m}}$. Locate the COM of the rod relative to 0.

$$\begin{aligned} X_{\text{com}} &= \frac{\int_0^L x(2x-1) dx}{\int_0^L (2x-1) dx} = \frac{2 \int_0^L x^2 dx - \int_0^L x dx}{2 \int_0^L x dx - \int_0^L dx} = \frac{2 \frac{L^3}{3} - \frac{L^2}{2}}{2 \frac{L^2}{2} - L} \\ &= \frac{2 \cdot \frac{8}{3} - \frac{4}{2}}{4 - 2} = \frac{\frac{16}{3} - \frac{4}{2}}{2} = \dots \end{aligned}$$

Ex:

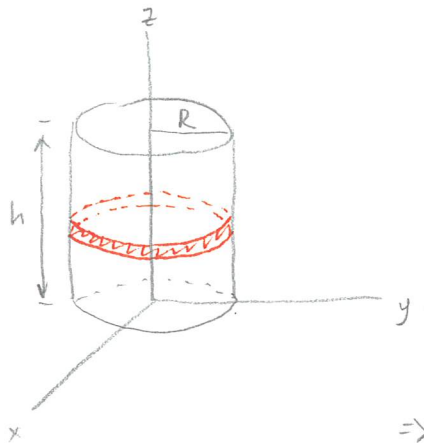


Find the COM of the rectangular plate shown in the figure. The plate has a uniform surface density σ .

$$\begin{aligned} X_{\text{com}} &= \frac{\int_0^a x dm}{M} = \frac{\int_0^a x b dx \sigma}{M} = \frac{\frac{\sigma}{ab} \int_0^a x b dx}{\frac{\sigma}{ab} \int_0^a \int_0^b dx dy} = \frac{1}{a} \frac{a^2}{2} = \frac{a}{2} \\ Y_{\text{com}} &= \frac{\int_0^b y dm}{M} = \frac{\int_0^b y a dy \sigma}{M} = \frac{\frac{\sigma}{ab} \int_0^b y a dy}{\frac{\sigma}{ab} \int_0^a \int_0^b dx dy} = \frac{1}{b} \frac{b^2}{2} = \frac{b}{2} \end{aligned}$$

$$\vec{R} = \frac{a}{2} \hat{i} + \frac{b}{2} \hat{j}$$

Ex:



Determine the COM of the cylindrical shell. It has a uniform surface density σ .

★ From the symmetry COM lies on the z-axis.

We divide the cylinder into very thin rings each of mass

$$dm = \underbrace{2\pi R dz}_{\text{surface area}} \sigma \quad \sigma = \frac{M}{2\pi R h}$$

$$\begin{aligned} \Rightarrow z_{cm} &= \frac{1}{M} \int_0^h z dm = \frac{1}{M} \int_0^h z 2\pi R dz \sigma = \frac{1}{M} \int_0^h z 2\pi R dz \frac{M}{2\pi R h} \\ &= \frac{1}{h} \int_0^h z dz = \frac{1}{h} \frac{h^2}{2} = \frac{h}{2} \end{aligned}$$

Velocity and acceleration of the COM

$$\vec{V}_{cm} = \frac{d\vec{r}_{cm}}{dt} = \frac{1}{M} \frac{d}{dt} \sum_i m_i \vec{r}_i = \frac{1}{M} \sum_i m_i \vec{v}_i = \frac{1}{M} \sum_i m_i \vec{v}_i$$

$\Rightarrow \vec{V}_{cm}$ and $\vec{a}_{cm}$ are weighted sums of individual velocities and accelerations like $\vec{R}$.

Similarly,

$$\vec{a}_{cm} = \frac{d\vec{V}_{cm}}{dt} = \frac{1}{M} \sum_i m_i \vec{a}_i$$

Momentum of a System of Particles

$$\vec{P}_{tot} = \sum_i \vec{p}_i = \sum_i m_i \vec{v}_i \Rightarrow \text{By using } \vec{V}_{cm} = \frac{1}{M} \sum_i m_i \vec{v}_i,$$

$$\Rightarrow \vec{P}_{tot} = M \vec{V}_{cm} \quad \text{linear} \quad \text{total momentum of the system is the linear momentum of the center of mass.}$$

Ex: Two particles of masses $m_1 = 1\text{kg}$ and $m_2 = 2\text{kg}$ have position vectors given by $\vec{r}_1 = (2t\hat{i} - 4\hat{j})\text{m}$ and $\vec{r}_2 = (5t\hat{i} - 2t\hat{j})\text{m}$. Determine the velocity and the linear momentum of COM at any time and at $t=1$.

$$\vec{r}_{cm} = \frac{\sum m_i \vec{r}_i}{\sum m_i} = \frac{1}{3} [12t\hat{i} + (-4 - 4t)\hat{j}] = \left[4t\hat{i} - \frac{4}{3}(t+1)\hat{j} \right] \text{m}$$

$$\vec{V}_{cm} = \frac{d\vec{r}_{cm}}{dt} = \left(4\hat{i} - \frac{4}{3}\hat{j} \right) \frac{\text{m}}{\text{s}}$$

$$\vec{P}_{tot} = M \vec{V}_{cm} = (12\hat{i} - 4\hat{j}) \frac{\text{kg m}}{\text{s}}$$

o At $t=1$:

$$\vec{r}_{cm} = \left(4\hat{i} - \frac{8}{3}\hat{j} \right) \text{m}$$

$$\vec{V}_{cm} = \left(4\hat{i} - \frac{4}{3}\hat{j} \right) \frac{\text{m}}{\text{s}}$$

$$\vec{P}_{tot} = (12\hat{i} - 4\hat{j}) \frac{\text{kg m}}{\text{s}}$$

Motion of System of Particles:

Remember: $\vec{a}_{cm} = \frac{1}{M} \sum_i m_i \vec{a}_i$

$\underbrace{\sum_i m_i \vec{a}_i}_{\vec{F}_i}$

Therefore Newton's 2nd law can be written as (for a system of particles)

$$\vec{a}_{cm} = \frac{1}{M} \sum_i \vec{F}_i$$

→ Force acting on the i th particle.

$\sum_i \vec{F}_i = M \vec{a}_{cm}$
total force acting on the system accelerates its com.

We can write $\vec{F}_i$ in terms of external forces and internal forces:

$$\vec{F}_i = \vec{F}_{i(ext)} + \sum_j \vec{f}_{ij}$$

we define the force in this way to put a boundary between the system and the rest of the universe and to understand how the system affects itself and how it is affected by the rest of the universe separately when in terms of our previous definitions

$\vec{F}_{i(ext)}$: resultant external force acting on i th particle.

Forces arising from factors other than other particles in the system.

e.g. think of a closed box containing n particles colliding with each other and the box is also in a gravitational field. The gravitational field acts as an external force whereas the forces applied on the particles because of collisions are internal forces.

$\vec{f}_{ij}$: Internal force exerted on the i th particle by the j th particle.

It is assumed that no force is applied on the particle by itself $\vec{f}_{ii} = 0$.

$$\vec{a}_{cm} = \frac{1}{M} \left(\sum_i \vec{F}_{i(ext)} + \sum_i \sum_j \vec{f}_{ij} \right)$$

But we know from Newton's 3rd law that $\vec{f}_{ij} = -\vec{f}_{ji}$. Hence, the $\sum_{ij} \vec{f}_{ij}$ term is zero and the net force acting on the system is due only to external forces.

$$\Rightarrow \vec{F}_{net} = \sum_i \vec{F}_{i(ext)} = M \vec{a}_{cm}$$

→ resultant external force on the center of mass. $\sum \vec{F}_{ext}$

By differentiating $\vec{P}_{tot} = M \vec{V}_{cm}$ with respect to time:

$$\frac{d}{dt} \vec{P}_{tot} = M \vec{a}_{cm} = \left(\sum \vec{F}_{ext} \right) \rightarrow \text{total ext. force acting on COM.}$$

★ Net external force acting on a system of particles is equal to the time rate of change of total linear momentum of the system

Conservation of Momentum:

If the system of particles is isolated from its surroundings, or in other words, the net external force is zero ($\sum \vec{F}_{\text{ext}} = 0$):

$$\sum \vec{F}_{\text{ext}} = 0 \Rightarrow \frac{d\vec{P}_{\text{tot}}}{dt} = 0$$

$$\Rightarrow \vec{P}_{\text{tot}} = M\vec{V}_{\text{cm}} = \text{constant}$$

* If no external force acting on a system of particles, the COM is either stay in rest or moves with a constant velocity.

Angular momentum of a System of Particles

The angular momentum of a system of particles about a fixed point is the vector sum of angular momenta of the individual particles:

$$\vec{L} = \vec{L}_1 + \vec{L}_2 + \dots + \vec{L}_n = \sum_i \vec{L}_i = \sum_i (\vec{r}_i \times \vec{p}_i) = \sum_i m_i (\vec{r}_i \times \vec{v}_i)$$

The Total Torque on a System

The total torque acting on an individual particle in a system is the sum of torques associated with the internal forces and external forces.

$$\begin{aligned}\vec{N}_i &= \vec{r}_i \times \vec{F}_i = \vec{r}_i \times \left(\vec{F}_{i,\text{ext}} + \sum_j \vec{f}_{ij} \right) \\ &= \vec{r}_i \times \vec{F}_{i,\text{ext}} + \sum_j \vec{r}_i \times \vec{f}_{ij}\end{aligned}$$

$$\sum_i \vec{N}_i = \sum_i \vec{r}_i \times \vec{F}_{i,\text{ext}} + \sum_{ij} \vec{r}_i \times \vec{f}_{ij} \rightarrow \text{For 2 particles: } \begin{array}{c} \vec{r}_1 \times \vec{f}_{12} + \vec{r}_2 \times \vec{f}_{21} \\ \vec{r}_1 \times \vec{f}_{21} + \vec{r}_2 \times \vec{f}_{12} \end{array}$$

We can write the term in double summation as:

$$\vec{r}_i \times \vec{f}_{ij} + \vec{r}_j \times \vec{f}_{ji} = (\vec{r}_i - \vec{r}_j) \times \vec{f}_{ij} \quad (\text{since } \vec{f}_{ij} = -\vec{f}_{ji})$$

- If we assume that the internal forces between the two particles lie along the line joining the particles, or equivalently, if we assume that the internal forces are central $\Rightarrow \vec{r}_i - \vec{r}_j \parallel \vec{f}_{ij} \Rightarrow (\vec{r}_i - \vec{r}_j) \times \vec{f}_{ij} = 0$

$$\vec{N}_{\text{tot}} = \sum_i \vec{N}_i = \sum_i \vec{r}_i \times \vec{F}_{i,\text{ext}}$$

$\Rightarrow$ Therefore, the total torque on the system about the origin is only the force torque associated with external forces:

$$\vec{N}_{\text{tot}} = \sum_{\text{ext}} \vec{r}_i \times \vec{F}_{i,\text{(ext)}}$$

The angular momentum and the total external torque

The angular momentum of individual particles may change with time. This will change the total angular momentum of the system.

$$\frac{d\vec{L}}{dt} = \sum_i \frac{d\vec{L}_i}{dt}$$

Consider the eq. we derived for torque:

$$\vec{N}_{\text{net}} = \sum_i \vec{r}_i \times \vec{F}_{i,\text{ext}}$$

$$= \sum_i \vec{r}_i \times \frac{d}{dt}(m_i \vec{v}_i) = \sum_i m_i \frac{d}{dt}(\vec{r}_i \times \vec{v}_i)$$

NOTE THAT : $\frac{d}{dt}(\vec{r}_i \times \vec{v}_i) = \underbrace{\frac{d\vec{r}_i}{dt} \times \vec{v}_i + \vec{r}_i \times \frac{d\vec{v}_i}{dt}}_0 = \vec{r}_i \times \frac{d\vec{v}_i}{dt}$

$$\Rightarrow \vec{N}_{\text{net}} = \frac{d}{dt} \left\{ \sum_i m_i (\vec{r}_i \times \vec{v}_i) \right\} = \frac{d}{dt} \left\{ \sum_i \vec{L}_i \right\} = \frac{d(\vec{L})}{dt}$$
$$= \sum \vec{N}_{\text{ext}}$$

* The net ^{external} torque acting about some origin exerted on a system of particles is equal to the time rate of change of the total angular momentum of the system.

Conservation of Angular Momentum

$$N_{\text{net(external)}} = 0 \Rightarrow \vec{L} = \text{const.} \quad \text{or} \quad \vec{L}_{\text{ini}} = \vec{L}_{\text{fin}}$$

Kinetic energy of a system of Particles

$$T = \frac{1}{2} \sum_{i=1}^n m_i v_i^2$$

REMEMBER: If SOP forms a rigid body (the relative dist. b/w. particles do not change in time) then this equation can be written as

$$T = \frac{1}{2} I \omega^2 \quad \begin{array}{l} \text{angular velocity} \\ \rightarrow \text{rotational inertia} \end{array}$$

Work:

$$\vec{F}_i = \vec{F}_{i(\text{ext})} + \sum_j \vec{f}_{ij}$$

Total work done on an individual particle

$$W_{12}^{(i)} = \int \vec{F}_i \cdot d\vec{s}_i$$

$$I = \sum_i m_i r_i^2$$

Work-Energy theorem for a system of Particles

State ①

State ②

The total work done in moving a system from one state to another is

$$W_{12} = \sum_i \int_1^2 \vec{F}_i \cdot d\vec{s}_i$$

$$\Rightarrow W_{12} = \sum_i \int_1^2 \vec{F}_i \cdot \frac{d\vec{s}_i}{dt} dt = \sum_i \int_1^2 \vec{F}_i \cdot \vec{v}_i dt = \sum_i \int_1^2 \left(\vec{v}_i \cdot \frac{d(m_i \vec{v}_i)}{dt} \right) dt$$

$$\vec{v}_i \cdot \frac{d(m_i \vec{v}_i)}{dt} = \frac{1}{2} \frac{d}{dt} [m_i (\vec{v}_i \cdot \vec{v}_i)] = \frac{1}{2} \frac{d}{dt} (m_i v_i^2)$$

$$\Rightarrow W_{12} = \frac{1}{2} \sum_i \int_1^2 \frac{d}{dt} (m_i v_i^2) dt = \sum_i \int_1^2 d \left(\frac{1}{2} m_i v_i^2 \right)$$

$$= \sum_i \left. \frac{1}{2} m_i v_i^2 \right|_1^2 = K_2 - K_1$$

total K.E.
in the second
state

total K.E.
in the
first state.

Conservation of Total Energy of a system of particles (without proof)

If the internal and external forces are all conservative then the total energy of a system of particles is conserved.

$$T + U = \text{constant}$$

Motion Relative to COM

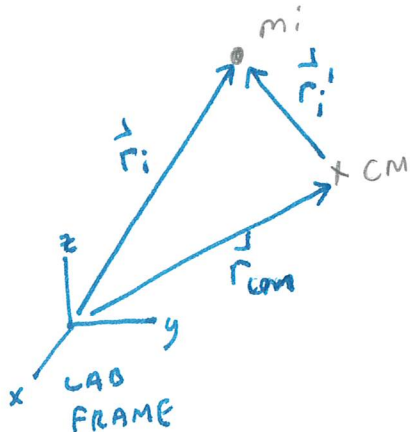
i) It simplifies the analysis.

ii) Adds a particular symmetry to the problem

$$\vec{r}_{cm} = \frac{\sum_i m_i \vec{r}_i}{M} \quad \text{w.r.t. LAB frame.}$$

$$\vec{r}_i' = \vec{r}_i - \vec{r}_{cm} \quad : \text{position vector of } i\text{th particle relative to COM.}$$

$$\vec{r}_i = \vec{r}_i' + \vec{r}_{cm} \quad : \text{position vector of } i\text{th particle relative to origin.}$$



- If we ^{put place} move the origin of our reference frame at the center of mass:

$$\vec{r}_{cm} = \frac{1}{M} \sum_i m_i \vec{r}_i \quad \vec{r}_{cm} = 0 \quad \sum_i m_i \vec{r}_i' = 0$$

$\vec{r}_i \leftrightarrow \vec{r}_i'$

The "distribution of mass of the particles about the CM is zero."

TOTAL LINEAR MOMENTUM:

- Therefore, the ~~total~~ linear momentum of the system is ~~zero~~ because

$$\sum_i m_i \vec{r}_i' = 0 \Rightarrow \sum_i m_i \frac{d}{dt} \vec{r}_i' = 0 \Rightarrow \sum_i m_i \vec{v}_i' = 0 \Rightarrow \sum_i \vec{p}_i' = 0 \Rightarrow \vec{p}' = 0$$

★ Total linear momentum of a system of particles is ZERO when the particles are observed from the center of mass.

TOTAL ANGULAR MOMENTUM:

$$\vec{r}_i = \vec{r}_i' + \vec{r}_{cm} \xrightarrow{d/dt} \vec{v}_i = \vec{v}_i' + \vec{v}_{cm}$$

$\vec{L} = \sum_i m_i (\vec{r}_i \times \vec{v}_i)$: total angular momentum

$\vec{v}_i$: velocity w.r.t LAB frame
 $\vec{v}_i'$: velocity w.r.t CM frame
 $\vec{v}_{cm}$: velocity of CM w.r.t LAB frame.

$$= \sum_i m_i [(\vec{r}_i' + \vec{r}_{cm}) \times (\vec{v}_i' + \vec{v}_{cm})]$$

$$= \sum_i m_i (\vec{r}_i' \times \vec{v}_i' + \vec{r}_i' \times \vec{v}_{cm} + \vec{r}_{cm} \times \vec{v}_i' + \vec{r}_{cm} \times \vec{v}_{cm})$$

$$= \sum_i m_i (\vec{r}_i' \times \vec{v}_i') + \underbrace{\left(\sum_i m_i \vec{r}_i' \right) \times \vec{v}_{cm}}_{0 \leftarrow \text{we have shown it previously} \rightarrow 0} + \vec{r}_{cm} \times \underbrace{\left(\sum_i m_i \vec{v}_i' \right)}_{0} + \left(\sum_i m_i \right) (\vec{r}_{cm} \times \vec{v}_{cm})$$

$$\vec{L} = \sum_i m_i (\vec{r}_i' \times \vec{v}_i') + M (\vec{r}_{cm} \times \vec{v}_{cm})$$

Linear momentum about CM

Linear mom. of CM about LAB frame.

TOTAL KINETIC ENERGY:

$$T = \frac{1}{2} \sum_i m_i v_i^2 : \text{Total w.r.t. LAB frame}$$

$$= \frac{1}{2} \sum_i m_i (\vec{v}_i \cdot \vec{v}_i) = \frac{1}{2} \sum_i m_i [(\vec{v}_i' + \vec{v}_{cm}) \cdot (\vec{v}_i' + \vec{v}_{cm})]$$

$$= \frac{1}{2} \sum_i m_i (\vec{v}_i' \cdot \vec{v}_i' + \vec{v}_i' \cdot \vec{v}_{cm} + \vec{v}_{cm} \cdot \vec{v}_i' + \vec{v}_{cm} \cdot \vec{v}_{cm})$$

$$= \frac{1}{2} \sum_i m_i v_i'^2 + \frac{1}{2} \sum_i m_i \vec{v}_{cm} \cdot \vec{v}_i' + \frac{1}{2} \sum_i m_i \vec{v}_i' \cdot \vec{v}_{cm} + \frac{1}{2} \sum_i m_i v_{cm}^2$$

$$= \frac{1}{2} \sum_i m_i v_i'^2 + \vec{v}_{cm} \cdot \underbrace{\left(\sum_i m_i \vec{v}_i' \right)}_0 + \frac{1}{2} \left(\sum_i m_i \right) v_{cm}^2 = \frac{1}{2} \sum_i m_i v_i'^2 + \frac{1}{2} M v_{cm}^2$$

Total K.E. about LAB frame is K.E. w.r.t CM + the K.E. of CM w.r.t. LAB frame.

TOTAL TORQUE ON A SOf about THE COM SYSTEM

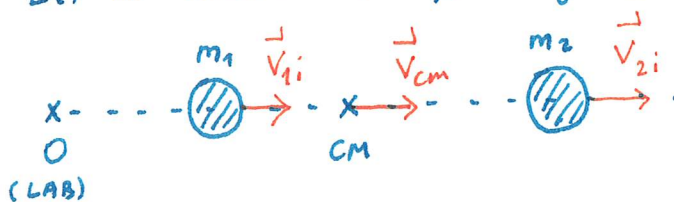
$$\vec{N}' = \frac{d\vec{L}'}{dt}$$

Collisions and the COM frame of reference

In problems involving collisions, it is useful to use an inertial frame of reference attached to COM to analyze the collision. This method is most commonly used in analysing collisions between subatomic particles or atoms. We know that

$$\vec{p}'_i = \vec{p}'_f = 0 \quad \text{from the point of view of COM frame.}$$

Let us consider the following 1D ^{case} problem:

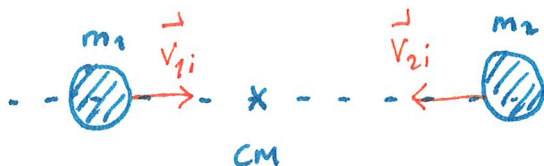


$$\vec{v}_{1i} > \vec{v}_{2i} \Rightarrow \text{The particles will collide.}$$

From the point of view of LAB frame:

$$\vec{p}_{\text{tot},i} = \vec{p}_{\text{tot},f} \Rightarrow m_1 v_{1i} + m_2 v_{2i} = m_1 v_{1f} + m_2 v_{2f}$$

From the point of view of CM frame:



$$\vec{p}'_i = \vec{p}'_f = 0$$

$$m_1 v'_{1i} + m_2 v'_{2i} = 0 \Rightarrow p'_{1i} = -p'_{2i}$$

$$m_1 v'_{1f} + m_2 v'_{2f} = 0 \Rightarrow p'_{1f} = -p'_{2f}$$

- when From CM frame, the two objects approach each other with equal and opposite momenta and move away from each other similarly.
- we need one less parameter to predict, say, v_{2f} in CM frame.

Ex: We have two particles moving in x-y plane.

$m_1 = 1 \text{ kg}$ $\vec{r}_1 = (t^2\hat{i} - 2t\hat{j}) \text{ m}$ → These are position vectors in LAB frame, where t is time.

Find: $m_2 = 2 \text{ kg}$ $\vec{r}_2 = (3t\hat{i} + \hat{j}) \text{ m}$

- Total angular momentum
 - Total external force torque
 - Total K.E
- Agree Angular momentum in CM frame
Total torque in CM frame
both in LAB frame and CM frame.
Total K.E. in CM frame

$$\vec{V}_1 = \frac{d\vec{r}_1}{dt} = (2t\hat{i} - 2\hat{j}) \text{ m/s}$$

$$\vec{V}_2 = \frac{d\vec{r}_2}{dt} = (3\hat{i}) \text{ m/s}$$

→ Total angular momentum in LAB frame:

$$\begin{aligned} \vec{L} &= \sum m_i (\vec{r}_i \times \vec{v}_i) = (1) [(t^2\hat{i} - 2t\hat{j}) \times (2t\hat{i} - 2\hat{j})] + (2) [(3t\hat{i} + \hat{j}) \times (3\hat{i})] \\ &= -2t^2\hat{k} + 4t^2\hat{k} + 2(\cancel{9t^2\hat{k}} - 3\hat{k}) = +2t^2\hat{k} - 6\hat{k} = (2t^2 - 6)\hat{k} \quad \text{kg} \frac{\text{m}^2}{\text{s}} \end{aligned}$$

→ Total K.E w.r.t LAB frame:

$$\begin{aligned} T &= \frac{1}{2} \sum_{i=1}^2 m_i v_i^2 = \frac{1}{2} (m_1 v_1^2 + m_2 v_2^2) \\ &= \frac{1}{2} [m_1 (4t^2 + 4) + m_2 9] = \frac{4}{2} (t^2 + 1) + \frac{18}{2} = 2(t^2 + 1) + 9 \end{aligned}$$

→ The net external torque about the origin:

$$\vec{N}_{\text{tot(ext)}} = \frac{d\vec{L}}{dt} = 4t\hat{k} \text{ (N.m)}$$

→ Total angular momentum w.r.t COM:

Remember: $\vec{L}_{\text{total}} = \left[\begin{array}{c} \text{linear mom.} \\ \text{about CM} \end{array} \right] + \left[\begin{array}{c} \text{linear mom. of} \\ \text{CM about LAB} \\ \text{frame} \end{array} \right]$

we calculated this This is asked we need to calculate this

$\vec{L}_1$

$$\vec{r}_{cm} = \frac{1}{M} \sum_i m_i \vec{r}_i = \frac{1}{3} [(t^2+6t)\hat{i} + (2-2t)\hat{j}] m$$

$$\vec{v}_{cm} = \frac{1}{3} [(2t+6)\hat{i} - 2\hat{j}] m/s$$

$$\vec{L}_{cm} = M (\vec{r}_{cm} \times \vec{v}_{cm}) = \frac{3}{9} \left\{ -(2t^2+12t)\hat{k} - \hat{k} [(2-2t)(2t+6)] \right\}$$

$$= -\frac{\hat{k}}{3} [2t^2+12t + 4t+12 - 4t^2-12t] = -\frac{\hat{k}}{3} [-2t^2+4t+12]$$

$$\vec{L}' = \vec{L}_{tot} - \vec{L}_{cm} = (2t^2-6)\hat{k} + \frac{\hat{k}}{3} (-2t^2+4t+12) = \left(2t^2 - \frac{2}{3}t^2 + \frac{4}{3}t + 4 - 6 \right) \hat{k}$$

$$= \left(\frac{4}{3}t^2 + \frac{4}{3}t - 2 \right) \hat{k} \text{ (J.s)}$$

→ ~~Not~~ ^{Total} external torque ~~about~~ ⁱⁿ com frame:

$$\vec{N}' = \frac{d\vec{L}'}{dt} = \left(\frac{8}{3}t + \frac{4}{3} \right) \hat{k} \text{ (J.s)}$$

→ Total K.E in com frame: K.E. of CM.

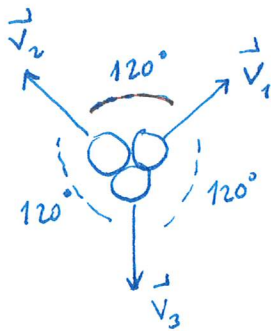
Remember: $\underbrace{T_{tot}}_{\substack{\text{Total K.E.} \\ \text{in LAB} \\ \text{frame} \\ \checkmark}} = T' + \underbrace{T_{cm}}_{\substack{\text{K.E. in CM} \\ \text{frame} \\ \text{? (if asked)}}}$
 ^{about LAB}
 ^{frame}
 ^(we need to calculate)

$$T_{cm} = \frac{1}{2} M v_{cm}^2 = \frac{1}{2} \frac{3}{9} [(2t+6)^2 + 4] J = \frac{1}{6} [4t^2 + 24t + 36 + 4] J$$

$$= \left(\frac{2}{3}t^2 + 4t + \frac{20}{3} \right) J$$

$$T' = T - T_{cm} = \dots$$

Ex:



A firework is launched vertically upward and explodes into three fragments of equal mass when it reaches the top of its flight at an altitude 40 m. If the two fragments land to the ground ~~after~~ 3s after the explosion, find the time it takes the third fragment to hit the ground. ($g = 10 \frac{m}{s^2}$)

KEY POINT: when the firework reaches the top its velocity before the explosion is zero

Conservation of momentum: $\vec{0} = m_1 \vec{V}_1 + m_2 \vec{V}_2 + m_3 \vec{V}_3$

$\downarrow$ $\downarrow$
 momentum before explosion momentum after explosion

Since $m_1 = m_2 = m_3 \rightarrow \vec{V}_1 + \vec{V}_2 + \vec{V}_3 = \vec{0}$ $2V = -V_3$ it is given

Since 1st and 2nd fragments land at the same time (t'), they must have the same vertical velocity at the moment of explosion. According to conservation of linear momentum their vertical velocities must be $-V_3/2$. Hence,

~~in given~~

~~For 1st and 2nd fragment:~~

$a = g$ $V_3 < 0$ $g < 0$
 $V = gt + V_0$
 $y = \frac{1}{2}gt^2 + V_0t + y_0$ given
 $\downarrow$
 the time when $y = 0$

For 3rd fragment:

given $(y_0) + V_3t + \frac{1}{2}gt^2 = 0 \rightarrow$

For 1st and 2nd fragment

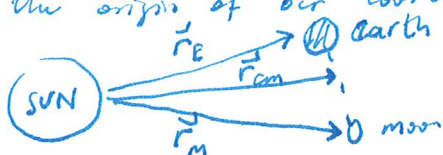
$y_0 - \frac{V_3}{2}t' + \frac{1}{2}gt'^2 = 0 \rightarrow y_0 + \frac{1}{2}gt'^2 = \frac{V_3}{2}t' \rightarrow 2\left(\frac{y_0}{t'} + \frac{1}{2}gt'\right) = V_3$

$(y_0) + 2\left(\frac{y_0}{t'} + \frac{1}{2}gt'\right)t + \left(\frac{1}{2}g\right)t^2 = 0$

$t_{1,2} = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$

Ex: Find the CM of Earth-Moon system and describe its motion around the sun.

We have seen that CM of Earth-Moon system traces and ellipse around sun. Since we can safely assume that there is no external force acting on the sun, we can assume that it is at least in an inertial frame of reference and at the origin of our coordinate system.



$\vec{r}_{cm} = \frac{M_E \vec{r}_E + M_M \vec{r}_M}{M_E + M_M}$

$\vec{F} = (M_E + M_M) \vec{a}_{cm}$
Eq. of motion of CM

(50-12)

The gravitational force ~~on Earth-Moon system~~ exerted by sun is

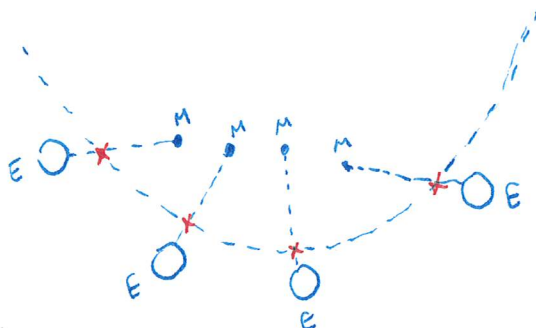
$$\vec{F} = -GM_s \left(\frac{M_E}{r_E^2} \hat{r}_E + \frac{M_M}{r_M^2} \hat{r}_M \right) \text{ where } \hat{r}_E \text{ and } \hat{r}_M \text{ are the unit vectors along } \vec{r}_E \text{ and } \vec{r}_M \text{ directions.}$$

Since the distance between earth and the moon is too small compared their distance from the sun, we may assume $r_E \approx r_M \approx r_{cm}$

$$\Rightarrow \vec{F} = -\frac{GM_s}{r_{cm}^2} (M_E + M_M) \hat{r}_{cm} = (M_E + M_M) \ddot{\vec{r}}_{cm}$$

$$\ddot{\vec{r}}_{cm} = -\frac{GM_s}{r_{cm}^2} \hat{r}_{cm}$$

Hence the CM of E-M system moves as a single planet of mass $(M_E + M_M)$ about the sun as below:

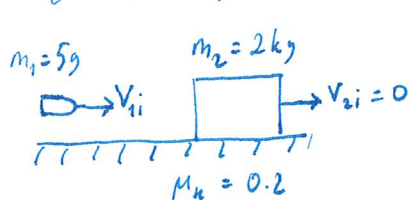


Collisions (Planned 29th of Nov. Wednesday)

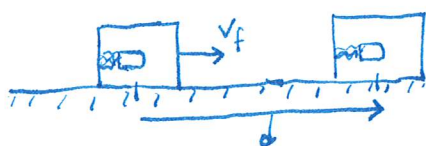
Elastic: K.E and momentum is conserved. (No energy loss)

Inelastic: K.E isn't conserved but the momentum is conserved. (Energy loss)
(Some or all K.E is transformed to heat energy.)

Ex: A $m_1 = 5g$ bullet is fired horizontally at the center of a wooden block with a mass $m_2 = 2kg$. The bullet embed itself in the block and the two slides a distance of $0.5m$ on a rough surface ($\mu_k = 0.2$) before coming to rest. Find the initial speed of bullet.



BEFORE



AFTER

$$P_i = P_f$$

$$m_1 v_{1i} + 0 = (m_1 + m_2) v_f$$

The energy loss during the movement of the body is equal to the work done by the friction:

$$\frac{1}{2} (m_1 + m_2) v_f^2 = F_f d = (m_1 + m_2) \mu_k d$$

$$\frac{v_f^2}{2} = g \mu_k d \Rightarrow v_f = \sqrt{2g \mu_k d}$$

$$v_{1i} = \frac{m_1 + m_2}{m_1} \sqrt{2g \mu_k d}$$

Ex: Two objects, A and B, with masses m_A and m_B respectively, move in a 2D space. Object A, with an initial velocity $\vec{v}_{Ai}$, is moving along the x-axis. Object B, ~~initially moving~~ has an initial velocity $\vec{v}_{Bi}$, along y-axis. After a perfectly inelastic collision, two objects stick together. Given their masses and initial velocities, determine:

- (i) The final velocity of the combined mass after collision.
- (ii) The direction of final velocity.